ETRO ELECTRONICS & INFORMATICS

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Analog Electronics Exercises

Session 1

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The CMOS table

The following table taken from the class lectures, chapter 1, summarizes the behaviour of small signal and large signal parameters given the inversion levels of a transistor. In the following exercises we will try to build intuition out of the different regions and the impact in the design of amplifiers.

inversion level	IDS – VGS dependence in saturation	gm – VGS dependence in saturation	VDSAT	Required W for given gm or IDS	gm/IDS	Main capacitors in saturation	fT	linearity
weak	exponential	exponential	Minimal (few times Ut)	Very high	Maximal (1/nUt)	1. Extrinsic Cgs and Cgd 2. Junction caps Csb and Cdb 3. Intrinsic Cgb	low	low
moderate	TRANSITION BETWEEN WEAK AND STRONG INVERSION							
strong	quadratic (linear with velocity saturation)	linear (constant with velocity saturation)	Larger than in weak inv.: Vov/a (long channel), less for short channel	low	Less than in weak inv.: 2/Vov 1/Vov with velocity saturation	1. Total Cgs 2. Junction caps Csb and Cdb 3. Extrinsic Cgd	high	higher

Figure 1: General table with MOS characteristics versus inversion level

Exercise 1

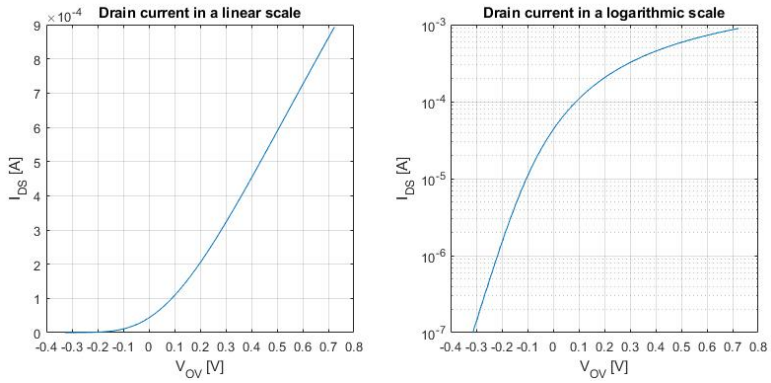


Figure 2: $I_{DS} \times V_{OV}$ curves (similar to Fig.1.26 from the class notes) showing the drain current per micrometer width ($W = 1\mu m$) as a function of V_{OV} on a linear scale (left) and logarithmic scale (right). The plots are generated for a $65nm$ nMOS transistor in saturation with $V_{DS} = 1V$, $V_{SB} = 0V$ and a gate length of $65nm$.

Given the I_{DS} curve in Fig.2 and the table in Fig.1, define the approximate range of V_{OV} for the three inversion regions: weak, moderate and strong.

a) Weak inversion:

b) Moderate inversion:

c) Strong inversion:

Exercise 2

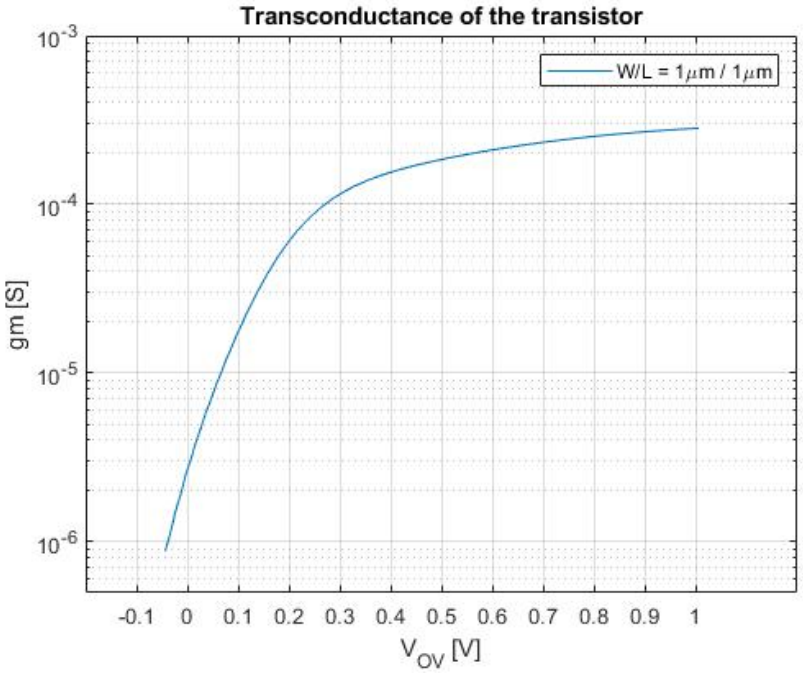


Figure 3: $g_m \times V_{OV}$ curve (similar to Fig.1.29 from the class notes) showing the transconductance per unit width of a long-channel nMOS transistor from a $65nm$ process as a function of V_{OV} . The transistor operates in saturation with $V_{DS} = 1.1V$, $V_{SB} = 0V$, $W = 1\mu m$, and $L = 1\mu m$.

Assuming we have a discrete component in the lab, which is a nMOS transistor in a $65nm$ technology with a W/L ratio of $1\mu m/1\mu m$ (Figure 3) and assuming the V_{th} of this nMOS is $0.3V$. How do we need to size V_{GS} to implement a transconductance of $0.1mS$? How to increase the g_m to $1mS$?
Hint: there are multiple answers. Cite as many as you can.

Exercise 3

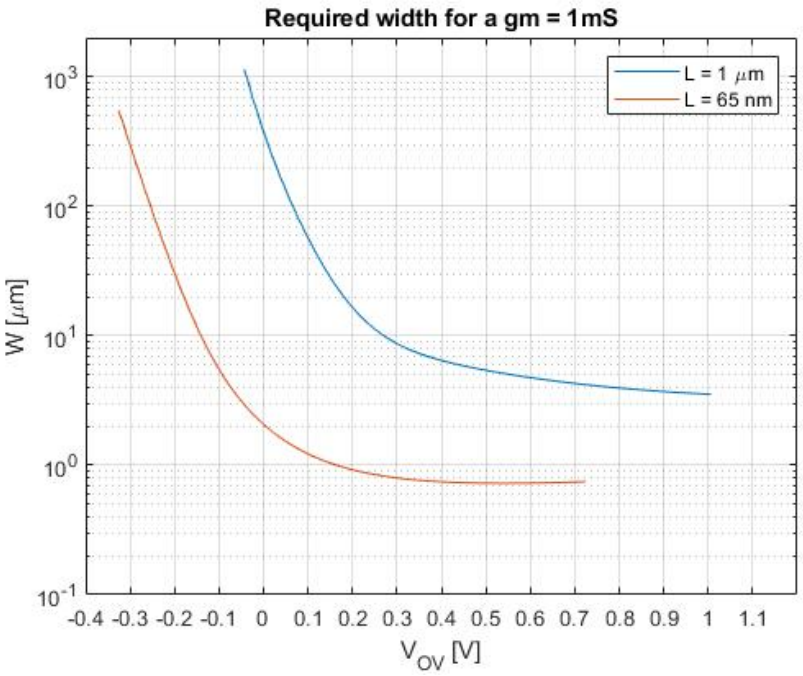


Figure 4: $W \times V_{OV}$ curve (similar to Fig.1.49 from the class notes) showing the required width as a function of the overdrive voltage for an nMOS transistor from a $65nm$ process with $V_{DS} = 1.1V$, $V_{SB} = 0V$, $L = 1\mu m$, and $L = 65nm$, to realize a transconductance of $1mS$.

Assuming now we can control on the W of the transistor and the L is fixed at $1\mu m$ such that we can derive the Figure 4. How can one approach to find the W to implement a $g_m = 10mS$ in the three inversion levels?
Hint: assume that $g_m = W \cdot f(ConstantParameters)$.

Exercise 4

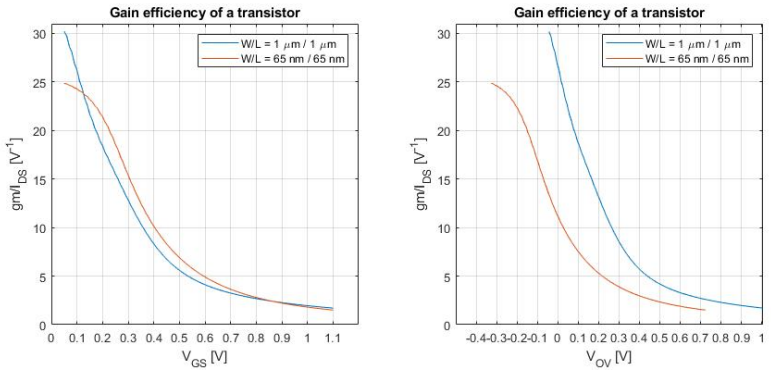


Figure 5: $g_m/I_{DS} \times V_{OV}$ curve (similar to Fig.1.30 from the class notes) showing the ratio g_m/I_{DS} for a transistor from a $65nm$ process with $V_{DS} = 1.1V$, $V_{SB} = 0V$, and two different channel lengths: $L = 1\mu m$ and $L = 65nm$.

a) What is the required I_{DS} current necessary to bias a $W/L = 1\mu m/1\mu m$ transistor with $g_m = 10mS$ at different inversion levels?

b) Assuming that this current is generated by a resistive load R_L, what is the voltage gain realized by the transistor?

Exercise 5

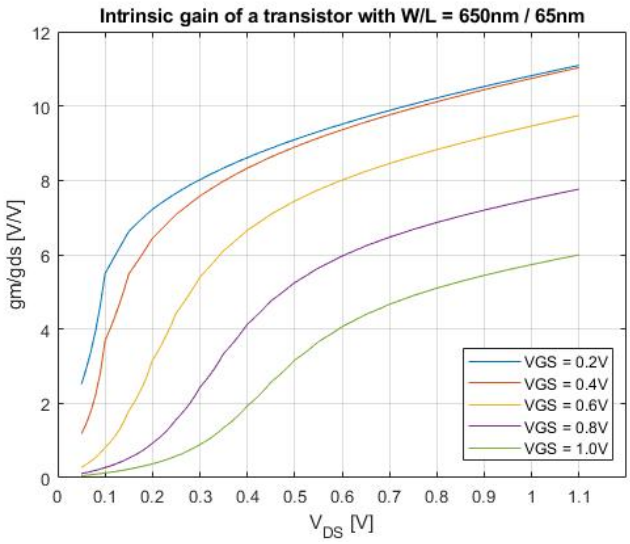


Figure 6: $g_m/g_{ds} \times V_{DS}$ curve (similar to Fig.1.36 from the class notes) showing the intrinsic gain as a function of V_{DS} for different V_{GS} values for a minimum-length transistor of a $65nm$ process with $V_{SB} = 0V$. Note that V_{th} is around $300mV$.

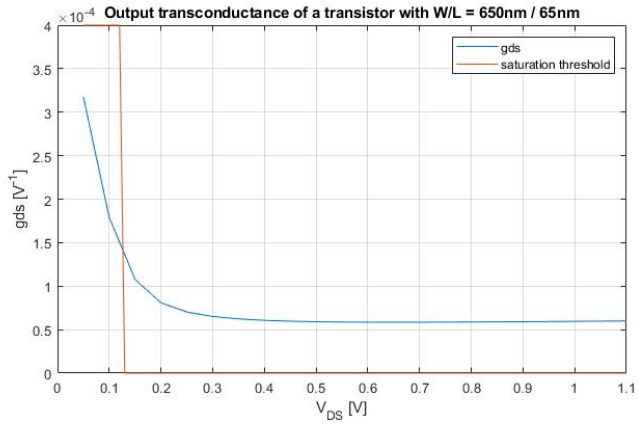


Figure 7: $g_{ds} \times V_{DS}$ curve (similar to Fig.1.33 from the class notes) showing the output conductance as a function of V_{DS} for a transistor in a $65nm$ process with $W = 650nm$, $L = 65nm$, $V_{GS} = 525mV$ (about $200mV$ above V_{th}), and $V_{SB} = 0V$.

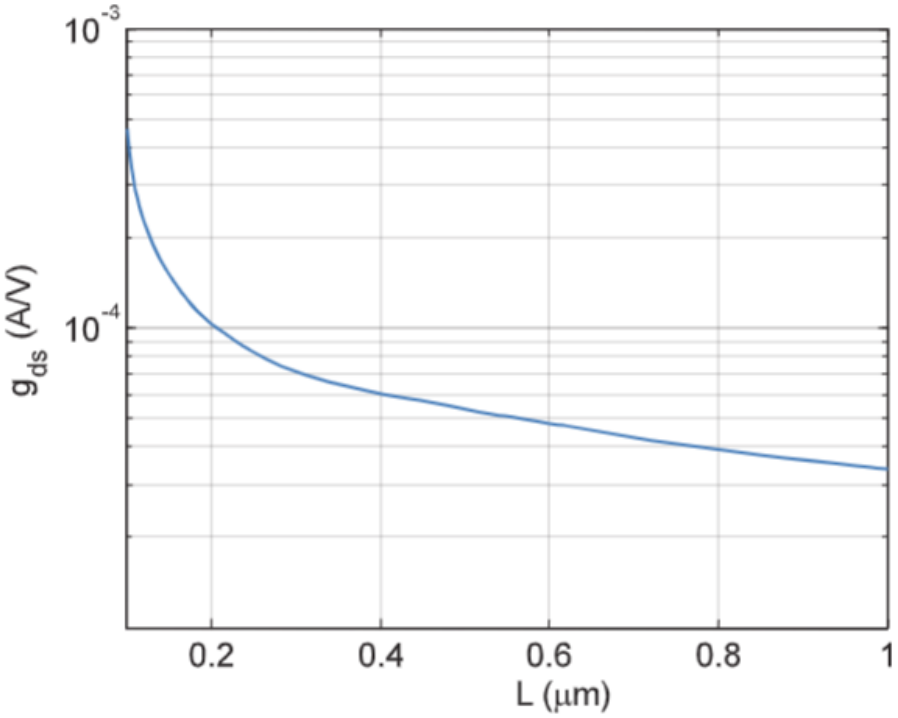


Figure 8: $g_{ds} \times L$ curve (similar to Fig.1.34 from the class notes) showing the output conductance as a function of channel length for a transistor in a $65nm$ process with $V_{OV} = 200mV$, $V_{DS} = 1.1V$, $V_{SB} = 0V$ and W is adapted such that W/L is constant and equal to 10.

Assuming now that this transistor is driven by a PMOS load as a current source and that the profiles of curves in Figure 7 and 8 are true for both PMOS and NMOS with the proper consideration of V_{DS} polarity: how does one chose the biasing conditions to maximize the gain of the common source amplifier?

Solution

Exercise 1

a) Weak Inversion: $V_{OV} < -0.1V...0.0V$

b) Moderate Inversion: $-0.1V...0.0V < V_{OV} < 0.2V...0.3V$

c) Strong Inversion: $0.2V...0.3V < V_{OV}$

Exercise 2

a) For $0.1mS$: bias the gate of the transistor with $V_{GS} = V_{OV} + V_{TH}$ at the point where $0.1mS$ is possible, so V_{OV} around $0.3V- > V_{GS} = 0.3V + 0.3V = 0.6V$

b) For $1mS$: it is not possible with only one transistor. One way to approach the problem is to add 10 transistors of $0.1mS$ in parallel with the same V_{GS} previously calculated, thus the equivalent transconductance will be $1mS$. Later in our design we will have control over W and L, thus there will be better ways to implement the transconductance by optimizing for other factors. Also, we must scale the current driving those transistors (by $10\times$). Other combinations ($100 \times 0.01mS$, $5 \times 0.2mS$) with the correct V_{GS} and bias current scaling are possible.

Exercise 3

We have to assume that $g_m = W \cdot f(CMOSParametersConstant)$, thus we are free to choose V_{OV}.

a) WI: $V_{OV} = -0.05V \rightarrow W = 1000\mu m@1mS \rightarrow W = 10000\mu m@10mS$

b) MI: $V_{OV} = 0.1V \rightarrow W = 50\mu m@1mS \rightarrow W = 500\mu m@10mS$

c) SI: $V_{OV} = 0.5V \rightarrow W = 5\mu m@1mS \rightarrow W = 50\mu m@10mS$

Exercise 4

a) Assuming the $10mS$ implementation from the previous exercise

   (a) WI: $V_{OV} = -0.05V \rightarrow g_m/I_{DS} = 31 \rightarrow I_{DS} = 322\mu A$
   (b) MI: $V_{OV} = 0.1V \rightarrow g_m/I_{DS} = 18 \rightarrow I_{DS} = 555\mu A$
   (c) SI: $V_{OV} = 0.5V \rightarrow g_m/I_{DS} = 4 \rightarrow I_{DS} = 2500\mu A$

b) The gain of the circuit will be defined as $A_v = g_m \cdot R_L$ assuming a small g_{ds} from the transistor. At the same time, the voltage at the output is defined as $V_{DD} - R_L \cdot I_{DS}$ and we need to define a safe value for it such that the transistor will be in saturation, let's check some cases assuming that $V_{DSAT} = V_{OV}$ for $V_{OV} = 0.1V$

   (a) $V_{DS} = 1.0V \rightarrow R_L = (1.1V - 1.0V)/555\mu A = 180\Omega \rightarrow A_v = 1.8V/V$

(b) $V_{DS} = 0.55V \rightarrow R_L = (1.1V - 0.55V)/555\mu A = 990\Omega \rightarrow A_v = 9.9V/V$

(c) $V_{DS} = 0.2V \rightarrow R_L = (1.1V - 0.2V)/555\mu A = 1.62k\Omega \rightarrow A_v = 16.2V/V$

(d) In general for this case: $g_m \cdot RL = g_m \cdot (V_{DD} - V_{DS})/I_{DS} = g_m/I_{DS} \cdot (V_{DD} - V_{DS})$. So the highest gain is achieved when V_{DS} approaches V_{DSAT}, but at what cost? Voltage swing and increased distortion of the signal.

Exercise 5

The gain of the PMOS loaded common source amplifier is given by $A_v = g_{m,NMOS}/(g_{ds,NMOS} + g_{ds,PMOS})$ where the PMOS can be optimized alone, but the NMOS has to have the ratio of transconductances (self gain) optimized.

a) PMOS: The gain is increased when $g_{ds,PMOS}$ is minimum. From figures 7 and 8, the transistor should be designed with large L and enough V_{DS} margin has to be given to keep this transistor in saturation. Usually a $V_{DS} = V_{DD}/2$ has a good compromise. The V_{GS} of this transistor is sized accordingly to drive the needed current to the NMOS branch. The inversion level will not have much effect in the base band gain, but it will have effect in the size W of the current source and this in the loading of the output node. Smaller size will be preferred in the future, thus a moderate to strong inversion should be chosen.

b) NMOS: assuming from the previous point that we reach a condition such that $g_{ds,NMOS} >> g_{ds,PMOS}$, the total gain of the structure is mainly dependant on the self gain of the NMOS (g_m/g_{ds} ratio). From figure 6 and assuming a $V_{DS} = 0.55V$ is desired, one can maximize the gain by biasing the transistor in weak inversion ($V_{GS} = 0.2V$), where the voltage gain will be around $14V/V$ or $23dB$. As we saw, this will lead to a big transistor size which will load the output node, so in case more bandwidth is needed, the designer will have to sacrifice some base band gain.